Hans William Alexander Hanley 汉斯 · 汉隶

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Research Interests

AI safety and truthfulness; disinformation and misinformation; network analysis; natural language processing; multilingual representation learning; computational social science.

Education

Ph.D., Computer Science, Stanford University 2025
GPA 4.016. Advisor: Professor Zakir Durumeric
Thesis: Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets. Meta PhD Research Fellow (Computational Social Science); National Science Foundation Graduate Fellow; Stanford EDGE Doctoral Fellow

M.S., Computer Science, Stanford University 2024
GPA 4.016. Advisor: Professor Zakir Durumeric
National Science Foundation Graduate Fellow

M.Sc., Statistical Science, University of Oxford 2020
Distinction (Highest Honours). Advisor: Professor Garrett Morris
Daniel M. Sachs Class of 1960 Graduating Scholarship, Worcester College

M.Sc., Advanced Computer Science, University of Oxford 2019
Distinction (Highest Honours). Advisors: Dr. Riccardo Spolaor, Professor Ivan Martinovic
Daniel M. Sachs Class of 1960 Graduating Scholarship, Worcester College

B.S.E., Electrical Engineering, Princeton University 2018
Highest Honours. GPA 3.982. Advisor: Professor Prateek Mittal
Concentration: Information Security and Privacy. Minors: Applications of Computing; Robotics and Intelligent Systems. Phi Beta Kappa; Tau Beta Pi; Sigma Xi

Professional Experience

Member of Technical Staff Apr. 2026–present
Microsoft AI, New York, NY

- Researched and developed rubric-based LLM-as-a-judge evaluators to assess human-AI conversations in Microsoft Copilot, measuring task completion, user intent satisfaction, reasoning complexity, response quality, and safety failures to inform model, system prompt, and product improvements.
- Developed LLM-based classifiers and measurement frameworks to connect real-world user feedback and behavior with offline model evaluations, identifying gaps in evaluation representativeness and failure-mode coverage; presented evaluation methodology and strategy to Microsoft AI's CEO.

Research Scientist, Central Applied Sciences Mar. 2025–Apr. 2026
Meta, New York, NY

- Researched supervised fine-tuning (SFT) techniques and persona-conditioned prompting to model individual behaviors and beliefs in large language models for digital twins, also leveraging verbalized sampling to increase response diversity and reduce behavioral homogenization.
- Developed and deployed large-scale multilingual and multimodal LLM pipelines for content understanding, creator classification, and evaluation of political content and integrity risks across Instagram and Threads to inform recommender systems.

Health AI Student Researcher Summer 2024
Google Research, Mountain View, CA

- Led research on how open-source LLMs such as Google's Gemma can be used as AI agents for running queries over, and understanding with high fidelity, structured health-related datasets.
- Built evaluation frameworks measuring accuracy, safety, and adversarial robustness for LLM-mediated medical data interactions, with verification layers to detect reasoning failures under distribution shift.

Digital Forensic Research Lab Research Intern Summer 2021
Atlantic Council, Washington, DC

- Researched and wrote seven reports for the DFRLab on cross-platform inauthentic behaviour, foreign influence operations, and propaganda narratives (listed under *Reporting*).

Software Engineering Intern, Business Applications Group (Data Engine) Summer 2018
Microsoft, Seattle, WA

- Implemented a new C# framework for the mobile offline synchronisation tool, allowing it to scale to 2× the number of users while reducing synchronisation time by 75%.
- Validated the framework with seven load and stress tests across a 30-server scale group.
- Created a concurrent row-versioning feature across multiple servers for more efficient data retrieval.

Software Engineering Intern, Gmail Security Summer 2017
Google, Sunnyvale, CA

- Designed and implemented a security feature for Gmail Android analysing outgoing email addresses to reduce user vulnerability.
- Implemented client security logic reducing request latency and bandwidth when processing recipient addresses against the Gmail backend.
- Refactored Gmail frontend and backend code analysing outgoing recipient addresses to improve reliability and readability.

Engineering Practicum Intern, Docs and Drive Security Summer 2016
Google, New York, NY

- Implemented an action executor in Java for the Google Sheets API v4 for randomised end-to-end security software, expanding authorisation test coverage for upcoming product launches.
- Contributed 52 changes to the Google codebase, including 37 scripted unit tests and 55 integration tests across 11 new Sheets API features.
- Presented the project to the Docs and Drive team in a 25-minute talk on features added, technical issues, and solutions.

Undergraduate Researcher, NSF Research Experience for Undergraduates Summer 2015
Texas Tech University, Lubbock, TX

- Contributed to the Texas Tech Answer Set Prolog (ASP) declarative programming codebase, exploring applications to AI and cybersecurity.
- Designed an ASP program using the Generate-and-Test methodology to identify de-anonymisation inference leakage in a k-anonymity framework.
- Presented a poster on ASP's functionality with Dr. Rattikorn Hewett.

Software Engineering Intern Summer 2014
Top Gun Laparoscopic Internship, Celebration, FL

- Wrote the engineering dossier for the Top Gun training instrument and presented it to the Chief Medical Officer of Lexion Medical.
- Built three Java programs with a Wiimote interface to teach laparoscopic surgeons and internists ultrasound technique; presented to Phillips Medical Systems for approval.

Peer-Reviewed Publications

1. P. Gerard, H. W. A. Hanley, L. Luceri, and E. Ferrara. *Bridging the Narrative Divide: Cross-Platform Discourse Networks in Fragmented Ecosystems*. 20th International AAAI Conference on Web and Social Media (ICWSM 2026). [link]
2. H. W. A. Hanley and Z. Durumeric. *Hierarchical Level-Wise News Article Clustering via Multilingual Matryoshka Embeddings*. The 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025). [link]
3. H. W. A. Hanley and Z. Durumeric. *Sub-Standards and Mal-Practices: Misinformation's Role in Insular, Polarized, and Toxic Interactions on Reddit*. The 28th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2025). **Best Paper Honourable Mention** (top 4%). [link]
4. H. W. A. Hanley and Z. Durumeric. *Twits, Toxic Tweets, and Tribal Tendencies: Trends in Politically Polarized Posts on Twitter*. The 28th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2025). [link]
5. H. W. A. Hanley, E. Okabe, and Z. Durumeric. *Tracking the Takes and Trajectories of English-Language News Stories from Trustworthy and Worrisome Websites*. 34th USENIX Security Symposium (USENIX Security 2025), August 2025. [link]
6. H. W. A. Hanley, Y. Lu, and J. Pan. *Across the Firewall: Foreign Media's Role in Shaping Chinese Social Media Narratives on the Russo-Ukrainian War*. Proceedings of the National Academy of Sciences (PNAS), January 2025. [link]
7. H. W. A. Hanley and Z. Durumeric. *Machine-Made Media: Monitoring the Mobilization of Machine-Generated Articles on Misinformation and Mainstream News Websites*. 18th International AAAI Conference on Web and Social Media (ICWSM 2024), June 2024. [link]
8. H. W. A. Hanley and Z. Durumeric. *Partial Mobilization: Tracking Multilingual Information Flows Amongst Russian Media Outlets and Telegram*. 18th International AAAI Conference on Web and Social Media (ICWSM 2024), June 2024. [link]
9. H. W. A. Hanley, D. Kumar, and Z. Durumeric. *Specious Sites: Tracking the Spread and Sway of Spurious News Stories at Scale*. 45th IEEE Symposium on Security and Privacy (Oakland 2024), May 2024. [link]

10. H. W. A. Hanley and Z. Durumeric. *TATA: Stance Detection via Topic-Agnostic and Topic-Aware Embeddings*. The 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023), December 2023. [link]
11. H. W. A. Hanley, D. Kumar, and Z. Durumeric. *A Golden Age: Conspiracy Theories' Relationship with Misinformation Outlets, News Media, and the Wider Internet*. 26th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2023), October 2023. [link]
12. H. W. A. Hanley, D. Kumar, and Z. Durumeric. *Happenstance: Utilizing Semantic Search to Track Russian State Media Narratives about the Russo-Ukrainian War on Reddit*. 17th International AAAI Conference on Web and Social Media (ICWSM 2023), June 2023. Invited paper, American Political Science Association (APSA) full-paper panel, "Mechanisms of Domination?: State Sponsored Information and Media Ecosystems," September 2023. [link]
13. H. W. A. Hanley, D. Kumar, and Z. Durumeric. *"A Special Operation": A Quantitative Approach to Dissecting and Comparing Different Media Ecosystems' Coverage of the Russo-Ukrainian War*. 17th International AAAI Conference on Web and Social Media (ICWSM 2023), June 2023. [link]
14. H. W. A. Hanley, D. Kumar, and Z. Durumeric. *No Calm in the Storm: Investigating QAnon Website Relationships*. 16th International AAAI Conference on Web and Social Media (ICWSM 2022), June 2022. [link]
15. H. W. A. Hanley, Y. Sun, S. Wagh, and P. Mittal. *DPSelect: A Differential Privacy Based Guard Relay Selection Algorithm for Tor*. 19th Privacy Enhancing Technologies Symposium (PoPETs 2019), July 2019. [link]

Preprints and Work in Progress

1. H. W. A. Hanley, E. Okabe, and Z. Durumeric. *M-STANCE: Soft Contrastive Learning and Synthetic Translations for Multilingual Stance Detection*. Preprint. Releases a 54-language multilingual stance detection dataset.

Dissertations

Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets

H. W. A. Hanley and Z. Durumeric. Stanford University, Ph.D. dissertation, Computer Science, 2025

GENerateZ: Automatic De Novo Design of Anticancer Drugs using Transcriptomic Data, Genetic Algorithms, and Variational Autoencoders

H. Hanley and G. Morris. University of Oxford, M.Sc. thesis, Statistical Science, 2020

PROMISE: Provably RObust Malware detectIon uSing diffErential privacy

H. Hanley, R. Spolaor, and I. Martinovic. University of Oxford, M.Sc. thesis, Advanced Computer Science, 2019

DPSelect: A Differential Privacy Based Guard Relay Selection Algorithm for Tor

H. Hanley, Y. Sun, S. Wagh, and P. Mittal. Princeton University, B.S.E. thesis, Electrical Engineering, 2018

Invited Talks and Posters

Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets
IDeaS Center Seminar Series, Carnegie Mellon University. April 7, 2026. Pittsburgh, PA

Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets
Google DeepMind Research Talk. February 25, 2026. London, UK

Investigating Patterns of Influence and Propaganda across International News Outlets
Weizenbaum Institute for the Networked Society. November 24, 2025. Berlin, Germany

Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets
Workshop on The Future of Social Media Research, Oxford Internet Institute. November 19, 2025. Oxford, UK

Narrative Networks: Investigating Patterns of Influence and Propaganda across International News Outlets
UC Santa Cruz Security Seminar. October 8, 2025. Santa Cruz, CA

Tracking the Takes and Trajectories of News Narratives from Trustworthy and Worrisome Websites
University of California, San Diego, Rising Stars in Data Science Workshop. November 14, 2024. San Diego, CA

Tracking the Takes and Trajectories of News Narratives from Trustworthy and Worrisome Websites
University of Cambridge, Computer Laboratory Security Seminar. November 8, 2024. Cambridge, UK

Thoroughly Tracking the Takes and Trajectories of News Narratives from Trustworthy and Worrisome Websites
Stanford University, Trust and Safety Conference. September 26, 2024. Stanford, CA

Tracking the Takes and Trajectories of News Narratives from Trustworthy and Worrisome Websites
Meta Computational Social Science Group. August 20, 2024. San Jose, CA

Narratives of Foreign Media Ecosystems in Chinese Social Media Discussions of the Russo-Ukrainian War
10th International Conference on Computational Social Science (IC2S2). June 18, 2024. Philadelphia, PA

Tall and True Tales: Thoroughly Tracking the Takes and Trajectories of News Narratives from Trustworthy and Worrisome Websites
10th International Conference on Computational Social Science (IC2S2). June 18, 2024. Philadelphia, PA

Narratives of Foreign Media Ecosystems in Chinese Social Media Discussions of the Russo-Ukrainian War
74th Annual Conference of the International Communication Association. June 20, 2024. Gold Coast, Australia

The Future of Computational Message Science: Theoretical Advances, Computational Frontiers, and Grand Societal Challenges (panellist)
74th ICA Pre-Conference, Queensland University of Technology. June 19, 2024. Brisbane, Australia

Thoroughly Tracking the Takes and Trajectories of News Narratives from Trustworthy and

Worrisome Websites
University of Chicago, People and Tech Seminar. April 5, 2024. Chicago, IL

Thoroughly Tracking the Takes and Trajectories of News Narratives from Trustworthy and Worrisome Websites
New York University, Center for Social Media and Politics. February 14, 2024. New York, NY

Influence of Foreign and Domestic Media Ecosystems on Chinese Social Media
TADA: New Directions in Analyzing Text as Data, University of Massachusetts Amherst. November 9, 2023. Boston, MA

Narratives of Foreign Media Ecosystems in Chinese Social Media Discussions of the Russo-Ukrainian War
Politics and Computational Social Science (PaCSS) 2023, UCLA. August 30, 2023. Los Angeles, CA

Happenstance: Utilizing Semantic Search to Track Russian State Media Narratives about the Russo-Ukrainian War on Reddit
119th American Political Science Association Annual Meeting. September 1, 2023. Los Angeles, CA

Narratives of Foreign Media Ecosystems in Chinese Social Media Discussions of the Russo-Ukrainian War
40th Annual Meeting of the Society for Political Methodology. July 9, 2023. Stanford, CA

Online Information Flows and Ecosystems: Understanding the Role of Misinformation and AI-Generated Media
University of Oxford, Oxford Internet Institute. June 13, 2023. Oxford, UK

Online Information Flows and Ecosystems: Understanding the Role of Misinformation and AI-Generated Media
University College London Information Security Seminar. June 12, 2023. London, UK

Tracking the Influence of Russian State Media Narratives about the Russo-Ukrainian War
University of Florida. September 6, 2022. Gainesville, FL

Tracking the Influence of Russian State Media Narratives about the Russo-Ukrainian War
University of Central Florida. August 16, 2022. Orlando, FL

Happenstance: Utilizing Semantic Search to Track Russian State Media Narratives about the Russo-Ukrainian War on Reddit
Stanford Security Lunch. May 11, 2022. Stanford, CA

Methods in the Madness: From QAnon to COVID-19, Conspiracy Theories' Relationship with Misinformation, the News Media, and the Wider Internet
Stanford Computer Forum. April 6, 2022. Stanford, CA

No Calm in the Storm
Stanford Security Lunch. September 8, 2021. Stanford, CA

Why Graduate School? My Experience in Graduate School
Princeton University. June 29, 2021. Princeton, NJ

GENerateZ: Automatic De Novo Design of Anticancer Drugs using Transcriptomic Data, Genetic Algorithms, and Variational Autoencoders
3rd RSC-BMCS/RSC-CICAG Artificial Intelligence in Chemistry. September 1, 2020. London, UK

An Exploration of Contextualized Word Vectors for Sentiment Analysis
Oxford Computer Science Conference. April 1, 2019. Oxford, UK

DPSelect: A Differential Privacy Based Guard Relay Selection Algorithm for Tor

Oxford Computer Science Conference. April 1, 2019. Oxford, UK

Reporting

All published by the Atlantic Council's Digital Forensic Research Lab (DFRLab).

- New COVID-19 conspiracy theories gain traction as old ones re-emerge
- Misinformation regarding France's COVID-19 "health passes" spread on Twitter
- Iranian social media exploit US-based conspiracies regarding USS Liberty incident (with A. Kann)
- China weaponizes discovery of graves at Canadian residential schools to avoid Xinjiang criticism
- Pro-Kremlin media echo Chinese outlets that COVID lab leak theory is a U.S. disinfo campaign
- Iranian accounts promote anti-Israel hashtags in the wake of the Israel-Hamas war
- Nigerian Twitter users flock to VPNs amid dubious legal threats (with J. Le Roux; Disinfo Africa and DFRLab)

Teaching

Teaching Assistant, CS265: Randomized Algorithms and Probabilistic Analysis Fall 2023
Stanford University, Professor Greg Valiant

- Assisted with instruction and grading; held weekly office hours.

Teaching Assistant, CS224N: Natural Language Processing with Deep Learning Winter 2023
Stanford University, Professor Chris Manning

- Led and helped design the default final project for the course.
- Supervised final-project implementation for 21 student teams.

Engineering Learning Consultant Fall 2020–Winter 2022
Stanford Center for Teaching and Learning

- Designed and implemented a matching algorithm for the peer buddy accountability program.
- Created and presented eight workshops for graduate students on Git, data presentation, R, and statistical methodology.
- Built a D3 tool to log and visualise student attendance at workshops and study halls.

Teaching Assistant, ELE 302: Building Real Systems Spring 2018
Princeton University, Professors Jeffrey Thompson and Stephen Lyon

- Guided students five hours weekly through hardware and software projects, including a PID speed-controlled car and autonomous car control.

Head Tutor, McGraw Center for Teaching and Learning Spring 2015–Spring 2018
Princeton University

- Tutored Physics (Mechanics and E&M) and Linear Algebra 6.5 hours weekly; ran trainings for undergraduate physics tutors.

Teaching Assistant, ELE 206: Contemporary Logic Design Fall 2016, Fall 2017
Princeton University, Professor Sharad Malik

- Taught and debugged Verilog in twice-weekly two-hour sessions; projects included Simon and a 16-instruction RISC ISA. Guided FPGA hardware programming sessions.

Instructor, Princeton Splash Spring 2017, Spring 2018
"Security and Privacy in the Digital Age" for New Jersey high-school students

Service, Mentorship, and Volunteering

Program committees

- AAAI — 2025, 2026, 2027
- AAAI-AISI (AI for Social Impact) — 2025, 2026, 2027
- IEEE Symposium on Security and Privacy — 2026
- ICWSM — 2023, 2024, 2025
- Poster Program Committee Associate Chair, ACM CSCW — 2024

Reviewing

- ACM CSCW — 2022, 2023, 2024, 2025
- ICWSM — 2023, 2024
- IC2S2 — 2024, 2025
- ACL — 2025
- NLP4Democracy — 2025
- International Communication Association (ICA) — 2024
- Stanford Journal of Science, Technology, and Society — 2022

Mentorship

- Graduate Mentor, Stanford CS Research Mentorship (2023–present) — mentee: Emily Okabe
- Graduate Mentor, Stanford CS Graduate–Undergraduate Mentorship Program (2022–2023) — mentee: Jenny Duan
- Mentor, Stanford CURIS summer undergraduate research program (2022) — mentee: Jaylene Martinez
- Graduate Mentor, Stanford CS Graduate–Undergraduate Mentorship Program (2021–2022) — mentee: Jaehyon Park
- Mentor, Stanford Undergraduate Research Association (SURA) (2021) — mentee: Laura Jean Bauman
- Interviewer, Daniel M. Sachs Global Scholarship (2021)
- Mentor, Princeton College Counseling, Pace Center for Civic Engagement (2014–2016)

Leadership

- President, National Society of Black Engineers, Princeton University chapter (2016–2017) — organised events with Google, Deloitte, Airbnb, Microsoft, McKinsey, and Capital One; built a mentorship program for underclassmen; designed nsbe.princeton.edu
- BAME Representative, Worcester College Middle Common Room, University of Oxford (2019–2020)
- BSE Freshman Interactor, Princeton University (2017–2018)
- Class of 2018 Representative, IEEE Princeton (2016–2017)

Selected Honours and Awards

Best Paper Honourable Mention, ACM CSCW 2025
Top 4% of submissions. Announcement

Rising Stars in Data Science, 2024
Hosted by UC San Diego with the University of Chicago and Stanford; celebrates and fast-tracks exceptional data scientists at the transition to postdoctoral, research scientist, or tenure-track positions. Details

AAAI/SIGAI Doctoral Consortium, 2025
Selected for the AAAI-25 Doctoral Consortium

Meta Research PhD Fellowship in Computational Social Science, 2023
Awarded to PhD candidates conducting research on emerging topics across computer science, engineering, and behavioural science; full tuition and fees for up to two years plus a $42,000 stipend. Announcement

Stanford Enhancing Diversity in Graduate Education (EDGE) Doctoral Fellowship
Offered to admitted Stanford doctoral students who contribute to the diversity of their field or degree program. Details

National Science Foundation Graduate Research Fellowship
Three years of funding and a stipend for university study and research at Stanford. Announcement

Daniel M. Sachs Class of 1960 Graduating Scholarship, Worcester College, University of Oxford
Assures two years' tenure and a stipend sufficient for university and college tuition at Oxford. Announcement

University of Oxford, Worcester College Prize for Academic Excellence
2019 and 2020

Hertz Foundation Graduate Fellowship Finalist, 2018
One of 40 candidates from a pool of 700 applicants selected as finalists after first-round interviews

Charles Ira Young Memorial Tablet and Medal, Princeton University
Awarded each year to the student who excels in research in Electrical Engineering. Announcement

James Hayes–Edgar Palmer Prize in Engineering, Princeton SEAS
Awarded annually to a senior who has manifested excellent scholarship, a marked capacity for leadership, and promise of creative achievement in engineering. Announcement

Phi Beta Kappa Society, Beta Chapter of New Jersey, Princeton University
Membership for exceptional academic achievement in the first three undergraduate years

Tau Beta Pi Engineering Honor Society
Oldest engineering honour society in the United States; membership awarded to the top eighth of the engineering class

Sigma Xi Scientific Research Society
Honour society for scientists and engineers; members are elected on the basis of research achievement or potential

George B. Wood Legacy Sophomore Prize, Princeton University
For the highest distinction and recognition in sophomore academic work. Announcement

Shapiro Award for Academic Excellence, Princeton University
2015 (freshman year) and 2016 (sophomore year)

McGraw Center for Teaching and Learning Excellent Tutor Award, 2016
For excellent tutoring during the 2015–2016 academic year

Rhodes Scholarship Finalist
One of 14 students in the Birmingham district invited to the final round of interviews

Marshall Scholarship Finalist
One of 20 students in the Atlanta district invited to the final round of interviews

Ron Brown CAPtain Scholar
Invitation-only professional development programme for exceptional African-American students; awarded to the top 3% of applicants

National Achievement Scholar for Academic Excellence, 2014

National Merit Scholar for Academic Excellence, 2014

Technical Skills and Languages

- **Proficient:** Python, Java, C, Go, R, Git, PyTorch, TensorFlow, Hugging Face
- **Experienced:** MATLAB, C#, C++, SQL, Answer Set Prolog (ASP)
- **Methods:** transformers; LLM fine-tuning; multilingual representation learning; interpretability and feature attribution; contrastive learning; LLM-based evaluation; safety and robustness analysis; network analysis
- **Languages:** English (native); Mandarin Chinese (HSK 5)